



Course Name

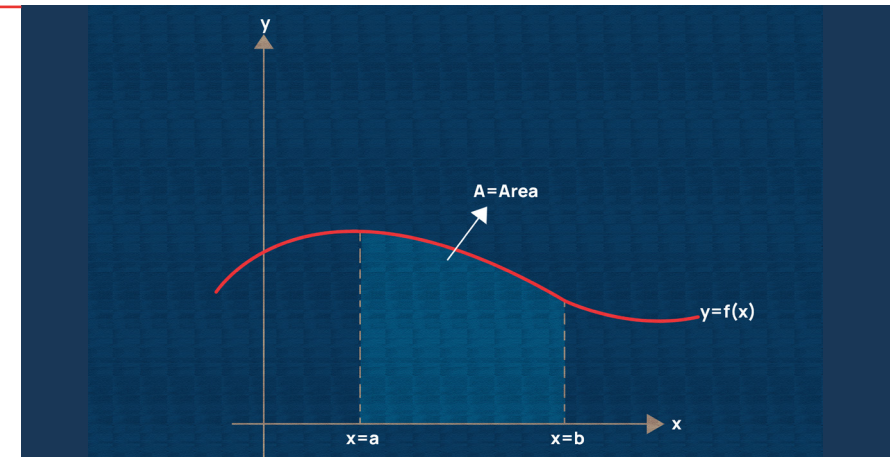
ELEMENTARY MATHEMATICS II (CALCULUS)

Topic: Methods of Integration

WEEK 8

IN SUMMARY

This week, you learned about integration techniques, a fundamental cornerstone of calculus. The lesson covered various methods of integration and the techniques required to apply them effectively. These techniques include integration by substitution, integration by parts, and partial fraction decomposition. Through comprehensive examples, you gained a deeper understanding of how to apply these methods to solve complex integrals.



BULLET POINT SUMMARY

- Integration is the process of finding the antiderivative of a function, which is the function whose derivative is the original function.
- Fundamental principles of integration:
- Constant multiple rule: The integral of a constant multiple of a function is equal to the constant multiple of the integral of the function.
- Sum rule: The integral of the sum of two functions is equal to the sum of the integrals of the two functions.
- Difference rule: The integral of the difference of two functions is equal to the difference of the integrals of the two functions.
- Substitution rule: This rule allows us to change the form of an integral by substituting a new variable for the original variable.

CASE STUDY

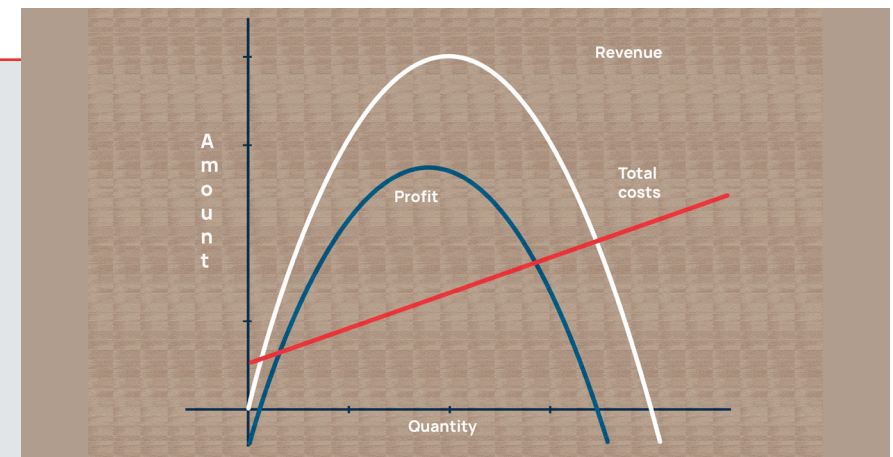
Title: Calculating Daily Sales Using Integration

Introduction

Integration is a fundamental concept in calculus with a wide range of applications in science and engineering. It is the process of finding the antiderivative of a function, which is the function whose derivative is the original function. Integration can be used to calculate the area under a curve, the volume of a solid, and the work done by a force.

Objective

This case study aims to demonstrate the application of integration concepts to solve a real-world



CASE STUDY CONT'D

problem in undergraduate studies. The specific problem that will be addressed is the calculation of the area under a curve that represents the daily sales of a product.

Methodology

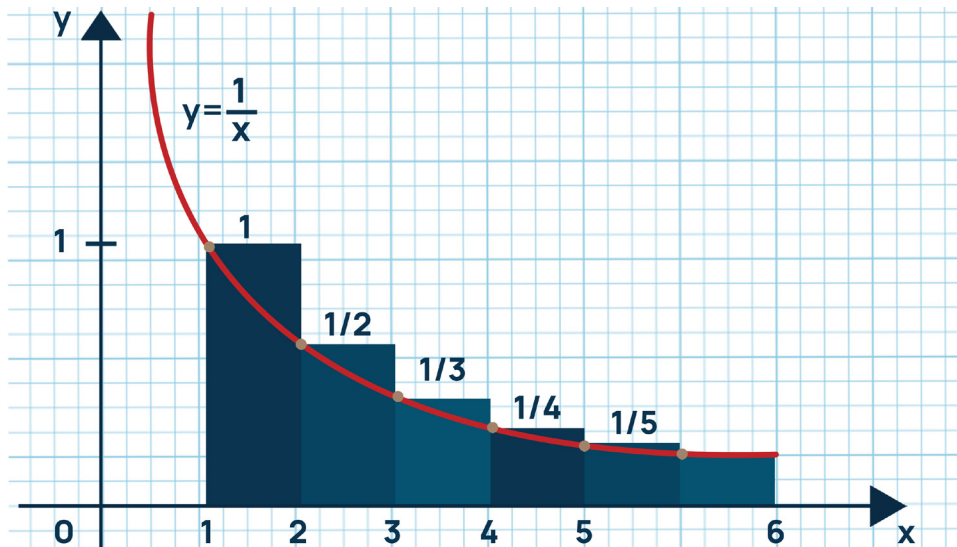
- The following steps will be used to solve the case study problem:
- Identify the function that represents the daily sales of the product.
- Find the antiderivative of the function using the appropriate integration technique.
- Evaluate the antiderivative at the desired time interval to calculate the area under the curve.

Results

The results of the case study will show how integration concepts can be used to solve a practical problem in undergraduate studies. The specific results will depend on the specific problem that is addressed.

QUESTIONS TO PONDER

- How can the integration of these techniques in engineering design be further optimised to enhance the efficiency and effectiveness of complex system analysis?
- In what ways can engineering education better emphasise the practical applications of integration techniques to prepare future engineers for real-world problem-solving challenges?



SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

- After this lesson, I have acquired skills in understanding integration techniques, applying mathematical rules, evaluating integrals, and using techniques like substitution, integration by parts, and partial fraction decomposition. Additionally, I have developed problem-solving and critical thinking abilities and gained insights into the practical application of these concepts in engineering.

PERSONAL REFLECTION

Studying integration techniques has deepened my understanding of the power of calculus in solving complex mathematical challenges. It has honed my critical thinking skills and highlighted the interdisciplinary nature of mathematical knowledge, particularly in its application to real-world problems.